



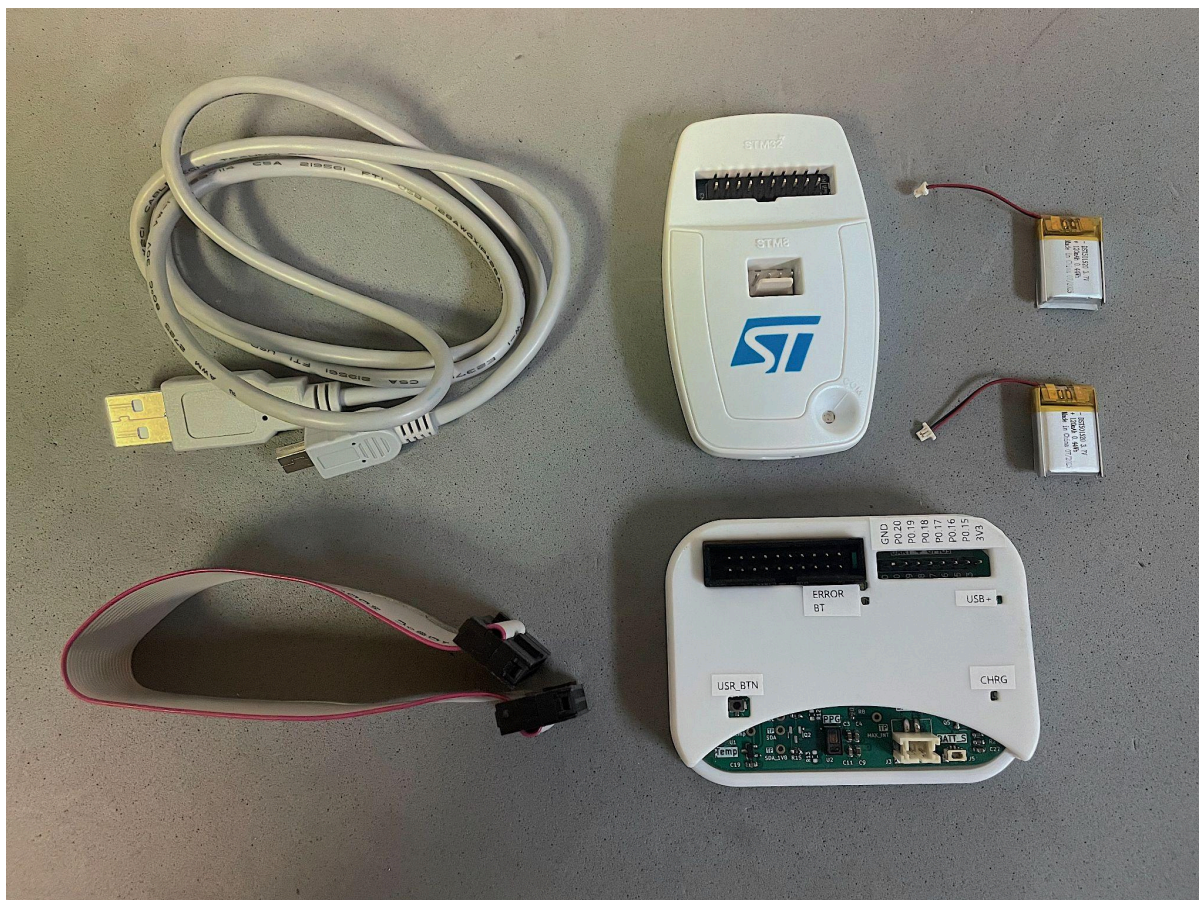
DocVoice Devboard Quick Guide

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Confidentiality Notice

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Devboard Kit Contents



Each Devboard kit contains the following:

1. 1 x USB-A to micro USB Cable
2. 1 x Ribbon Cable
3. 1 x ST-Link
4. 2 x Small LiPo 120mAh Batteries
5. 1 x DocVoice DevBoard

How to Program

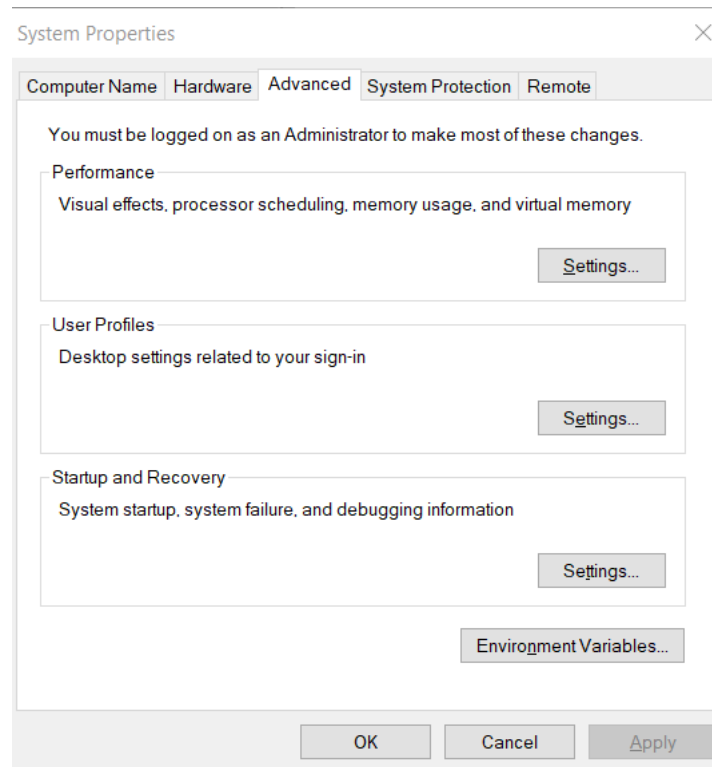
Installing Dependencies

In order to program the device you will need to download OpenOCD. OpenOCD is an open-source tool that will enable the ST-Link to flash firmware onto your device.

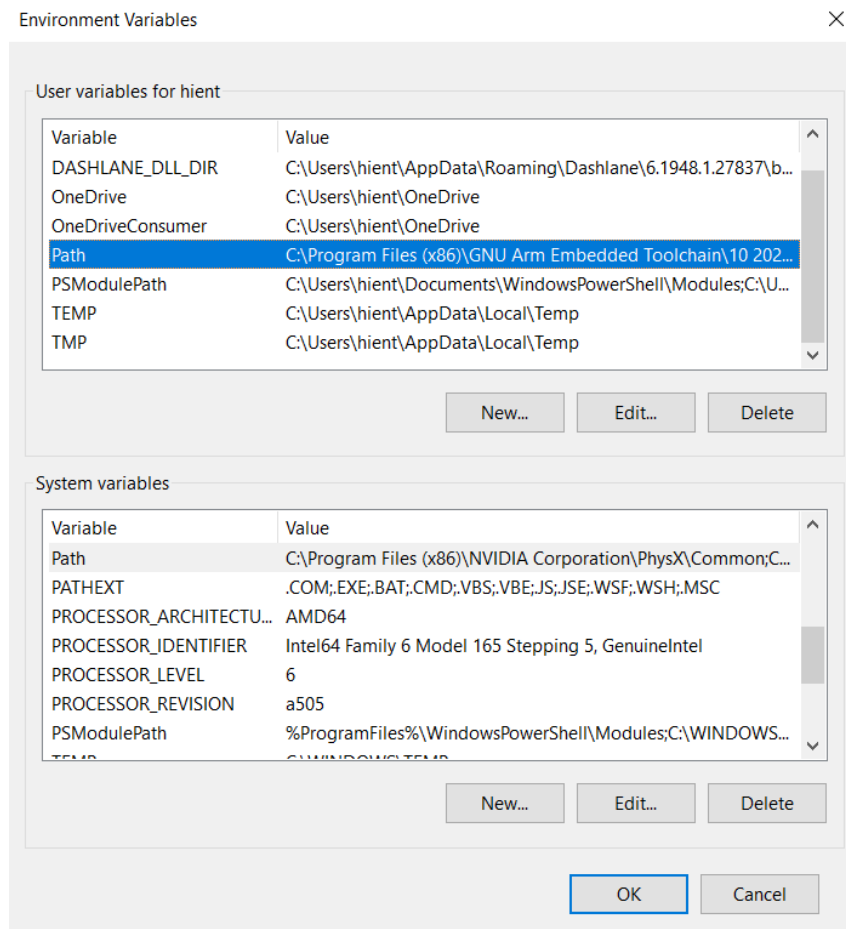
Windows:

Download the latest available version of openOCD from gnutoolchains.com (as of October 2025, the latest version is 20250710)

1. Unzip and place it directly into your C drive
2. Add the path "C:\openocd-version-number\bin" to your environment variable
3. Search in Windows search bar "Edit the system environment variables" to open System Properties panel



4. Go to Environment Variables



- Under User variables, choose Path and click Edit. Add the path "C:\openocd-version-number\bin" to your environment variable

Mac:

- For MacOS you can use brew: (Search up download homebrew if you haven't set it up yet).
`brew install openocd`
- That directory should already be in your PATH if Homebrew is set up correctly. You can confirm by running:
`which openocd`
- If it outputs something like `/opt/homebrew/bin/openocd` (Apple Silicon) or `/usr/local/bin/openocd` (Intel), then it's already in your PATH.

Installing ST-Link Driver

In order for windows to recognize the ST-Link, please install the following host drivers:

<https://www.st.com/en/development-tools/stsw-link009.html#st-get-software>

(Mac users can skip this step, as the OS already has the necessary USB class drivers from installing openOCD).

Connecting the Devboard

To set up the ST-Link for flashing, connect the USB-A end of the cable to your computer and the micro-USB end to the ST-Link programmer. Then attach the ribbon cable from the programmer to the programming header on your device. The ribbon connectors are keyed with notches, so you don't need to worry about misalignment.



Powering the Devboard

The development kit includes two batteries that can connect to the Batt_S connector located at the bottom right of the board. One was provided as a backup, it only needs one battery to operate, **never operate with more than one battery**. Make sure a battery is connected before attempting to flash the device.

Batteries can be recharged through the USB-C port on the side of the board. **Please note that the device should never be powered by the charger alone, a battery must always be connected during operation and flashing.** The device will still function normally with the charger connected along with a battery.

Note that the device automatically shuts down if the battery output is at 3.5V or lower to prevent the battery from being damaged.

Running Script Files

The device will arrive already programmed. It has been tested and should work out of the box. In the future, updates to the firmware may be necessary.

You will be provided both a command script file (`.command`) for macOS and a batch script (`.bat`) for Windows to automatically flash the firmware. These should be found in the folder named External Programmer. Open the script appropriate for your operating system, and it will carry out the firmware update process for you.

The scripts will prompt you for a Device ID number. This number can be found on a label on the back of the device (ex. 1 or 2).

Device Behaviour

Power and Startup

The device will immediately start advertising as soon as it is connected to the battery. After a period of inactivity (5 minutes), the device will then go to sleep. A button press must then be used to wake the device. It will immediately start advertising upon wakeup.

The sleep duration can be adjusted through the user settings, through bluetooth.

Button Functions

The button found on the left side of the devboard has two main functions.

Short Button press

- Will show the status of the battery.

Long Button Press (above 5 seconds)

- Factory reset - clear all data including BLE pairing data

Please be advised that a toothpick might be needed to press the button through the enclosure. Alternatively, the lid can be removed for better access.

Charging Behaviour

Charging is done through the USB-C. The device can still function while it is connected to the charger.

LED Indicators

The development board uses a red and a blue LED to indicate the following status information.

Battery Status

- Solid blue LED: Battery Level > 75%
- Fast blink blue LED: Battery Level is 50%-75%
- Solid red LED: Battery level is 25%-50%
- Fast Blink red LED: Battery level <25%

Sensor Error

The red light is also responsible for indicating sensor errors. Refer to the Developer Reference spreadsheet for more info. On when this might be encountered.

BLE Advertising

Blue LED blinks fast when its bluetooth is publicly advertising, and blinking slowly if privately advertising.

PPG Sensor

For best results, place your index finger on the PPG sensor. Once the app begins collecting data, the firmware will verify that the sensor detects finger contact. When a finger is detected, data will begin transmitting over BLE. If the finger is removed at any point during acquisition, transmission will pause and the system will wait until the finger is detected again before resuming.

Temp Sensor

The temp sensor is also exposed for testing purposes. The sampling rate can be adjusted through BLE configurations. Please see the Developer Reference Guide document for more information.

Data Retention

The device is designed to keep user/sensor settings between power cycles. A factory reset can be performed by holding the user button for at least 5 seconds. This will clear all data including BLE pairing data.

App Behaviour

A spreadsheet (Developer Reference Guide - DocVoice.xlsx) has been created to assist developers with commands and device states. The device relies on Bluetooth Low Energy (BLE) for wireless data transmission through GATT services. The sections below outline the content of these spread sheets.

App Dev Guide DocVoice

This device runs on a state machine that can be navigated through specific commands. This sheet lists the required steps to enter the different states through BLE commands. Read the notes left to troubleshoot issues as they arise.

Error Codes

The firmware implemented for the device has been configured to send an error code when necessary. These error codes are disclosed in this sheet, where the meaning of the value of the error can be deciphered. These error codes will grow in number once more features are implemented into the device. Review the BLE GATT Characteristics sheet for more information about how to read the error codes of the device.

BLE GATT Characteristics

Once the App Dev Guide is traversed correctly, the device will be prepared to send data through BLE. The characteristics are organized to correlate to specific uses. When connecting to the device and following the App Dev Guide, these characteristics will be shown. The properties of these characteristics are defined in this sheet such as UUID, read/write properties and others. Common uses of this sheet are reading the state of the device, determining the battery level, reading the PPG measurements, and more.

Device Settings Byte Structure

The device settings can be configured through BLE. These settings must be configured appropriately for your function. Key settings for the sensors and the device are listed in this sheet for convenient configuration through the Settings characteristic. All sensor configurations are for the PPG sensor for this device and other device settings are applicable for identification. Default values have been tested previously, however, feel free to configure them as you see fit for power consumption and other optimizations.

State Table

This sheet documents the possible states of the device which can be read through the State characteristic. To further understand how these states interact with each other, review the App Dev Guide sheet.

Hardware Reference Diagram

